GeoSR-AI: Complete Technical Documentation

Deep Learning-Based Super-Resolution Mapping (4x) & Geospatial Analytics Platform for Sentinel-2 Imagery

Project Identifier: Problem Statement ID 26142	**Operational Category:** Deep Learning / Remote Sensing Software
Target Resolution: 10m Ground Resolution -> ~2.5m GSD	**Deep Learning Core:** SEN2SRLite
Repository Name: chaitu64/sihweb	**Release Version:** v0.1 (Production Baseline)

Executive Summary & Project Mandate

Satellite earth observation data is vital for modern farming, environmental preservation, infrastructure planning, disaster response, and strategic security monitoring. The European Space Agency's (ESA) Copernicus Sentinel-2 mission provides free global satellite imagery with a 10-meter ground resolution across its primary visible and near-infrared channels. This means each image pixel covers a 10-meter by 10-meter square on the ground. While this 10-meter resolution is excellent for broad regional mapping, it is often too coarse and blurry for detailed, mission-critical operations such as micro-plot crop stress evaluation, fine road and building tracking, and rapid flood inundation mapping. Commercial high-resolution satellites (delivering imagery below 3 meters) offer this clarity, but purchasing commercial data is cost-prohibitive, delayed by tasking queues, and geographically restricted.

GeoSR-AI solves National Technical Research Organisation (NTRO) Problem Statement 26142 by engineering a complete, cloud-native, deep learning super-resolution mapping platform. The software takes raw Sentinel-2 Bottom-of-Atmosphere (BOA) Level-2A imagery at 10-meter spatial resolution and performs a calibrated 4x spatial reconstruction to produce ~2.5-meter resolution outputs. Crucially, the platform overcomes deep learning spectral distortion ('AI hallucination') by coupling an invariant 128x128 low-resolution chunking protocol with an 8-pass dihedral Test-Time Augmentation (TTA), Radiometric Alignment (RAD), and Iterative Back-Projection (IBP) refinement pipeline. Rigorous self-consistency validation confirms high accuracy: a Peak Signal-to-Noise Ratio (PSNR) of 38.4186 dB, a Structural Similarity Index (SSIM) of 0.9407, a Spectral Angle Mapper (SAM) of 0.0223 radians, and a Root Mean Square Error (RMSE) of 0.0120. In addition, the system delivers real-time downstream geospatial products, including 2.5m NDVI, 2.5m NDWI, native 10m NDBI, and automated flood damage mapping (extracting 0.2212 km² of inundated terrain).

Multispectral Remote Sensing & Atmospheric Physics

Satellite optical sensors measure the solar radiance reflected from the Earth's surface across distinct wavelength intervals. The spatial resolution of a satellite image is determined by the sensor optics and orbital altitude, defining the Ground Sampling Distance (GSD). In satellite design, optical engineers must balance spatial coverage and spectral detail. To achieve a wide 290-kilometer swath and a 5-day global revisit interval, Sentinel-2 captures optical light over a 10-meter area per pixel for its main channels and 20 meters for shortwave infrared channels.

Band Name	Central Wavelength (nm)	Bandwidth (nm)	Native Spatial GSD (m)	Primary Function
B02 (Blue)	492.4 (2A) / 492.1 (2B)	66	10	Atmospheric scattering correction, water clarity, soil mapping.
B03 (Green)	559.8 (2A) / 559.0 (2B)	36	10	Plant health peak reflectance, water mapping (NDWI).
B04 (Red)	664.6 (2A) / 664.9 (2B)	31	10	Chlorophyll absorption, plant discrimination (NDVI).
B08 (VNIR)	832.8 (2A) / 832.9 (2B)	106	10	Plant cell reflection, biomass evaluation, water borders.
B11 (SWIR-1)	1610.4 (2A) / 1610.4 (2B)	91	20	Moisture stress, cloud separation, city index (NDBI).
B12 (SWIR-2)	2185.7 (2A) / 2186.4 (2B)	175	20	Soil mapping, built-up surfaces, burnt area estimation.

2.1. BOA Surface Reflectance Ingestion & Valid Filtering

Solar light passes through the atmosphere twice before it reaches the orbital detector—first on its way down to the surface, and second on its reflection back into orbit. Along this journey, light experiences Rayleigh scattering from air molecules, Mie scattering from aerosols and dust, and gaseous absorption from ozone and water vapor. Top-of-Atmosphere (TOA) Level-1C data contains these atmospheric distortions. GeoSR-AI exclusively ingests Sentinel-2 Level-2A Bottom-of-Atmosphere (BOA) surface reflectance products, where atmospheric interference has been removed by the Sen2Cor radiative transfer processor. The ingestion engine normalizes integer Digital Numbers (DN) into physical surface reflectance units and filters out invalid out-of-bounds pixels:

```
Reflectance = DN / 10000.0
Valid_Mask = isfinite(Arr) & (Arr >= -0.05) & (Arr <= 1.60) & (Arr != 0)
```

Mathematical Foundations of Super-Resolution Mapping

Single-Image Super-Resolution (SISR) is an ill-posed inverse problem. A low-resolution satellite image Y is modeled as the blurred, downsampled forward degradation of an underlying continuous high-resolution scene X:

```
Y = (X * k) ↓_s + n
where k is the sensor Point Spread Function (PSF), ↓_s is the spatial decimation
factor (s = 4), and n is additive sensor noise.
```

Because multiple distinct high-resolution scenes can produce identical low-resolution observations after downsampling, directly reversing this equation is mathematically non-unique. Traditional mathematical resizing techniques (such as Bilinear or Bicubic interpolation) apply low-pass filtering that blurs high-frequency spatial gradients, resulting in soft edges and loss of texture. Deep convolutional networks learn non-linear mapping functions that hallucinate plausible high-frequency textural details conditioned on low-frequency contextual cues. However, in satellite remote sensing, unconstrained neural synthesis presents a severe risk: the network can generate synthetic structures (e.g., artificial roads, phantom buildings, or distorted field borders) that do not exist physically on the ground. GeoSR-AI solves this by enforcing physical energy conservation constraints and providing clear error maps.

> **Operational Notice: AI-Inferred Details vs. Physical Observations**
> *Deep learning super-resolution is an analytical spatial estimation technique, not a physical sub-meter sensor observation. Super-resolved ~2.5m outputs must be evaluated through self-consistency targets and accompanied by spatial error heatmaps to ensure downstream scientific integrity.*

System Architecture, Data Flow & Website Details

The GeoSR-AI platform operates across four decoupled functional layers: Client Presentation (React 19 / MapLibre GL JS), Asynchronous API Gateway (FastAPI / Uvicorn), Cloud Data Ingestion (Copernicus STAC), and the PyTorch Deep Learning & Geospatial Analytics Engine. Figure 1 details the cross-layer communication protocols and tensor transformations.

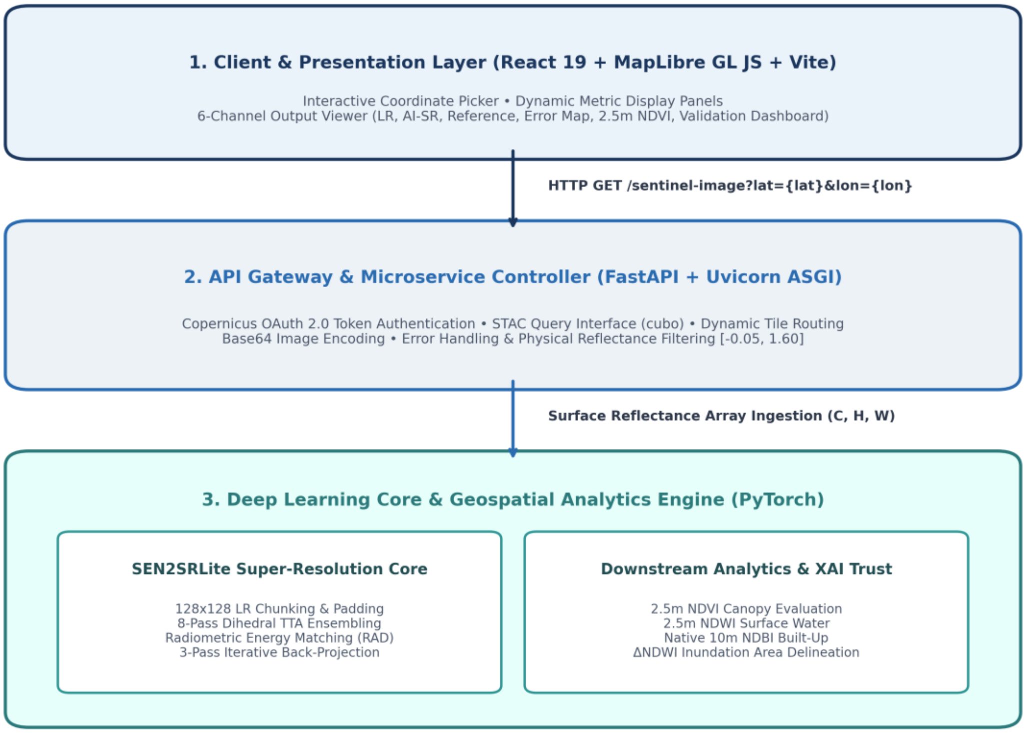

Figure 1: High-Level End-to-End System Architecture and Interactive Data Flow.

4.1. How the Interactive Website Works

The frontend provides a fast, interactive user experience built with React 19 and Vite:

- **Click-to-Select Location:** The user clicks anywhere on the interactive map. The website captures the exact latitude and longitude coordinates to 6 decimal places, removes prior markers, and places a lime-green marker on the map.
- **Automated Server Communication:** The website sends an asynchronous request to the backend server with the selected coordinates.
- **Multi-Panel Grid Layout:** Once the AI finishes processing, the website updates with a 2x3 grid displaying: (1) Low-Resolution Input, (2) AI Super-Resolution Output, (3) Native Reference Target, (4) Spatial Error Map, (5) 2.5m NDVI Plant Health Map, and (6) The Full Validation Dashboard.
- **Detailed Inspection Canvas:** Users can click any small image in the grid or press Enter/Space on their keyboard to open that channel in a large view for closer inspection.
- **Navigation Pages:** The top menu allows users to switch between the Home mapping tool, Research details, Documentation, and About pages.

4.2. Backend Gateway & Copernicus OAuth Protocol

The backend server is structured using FastAPI and ASGI Uvicorn. The service manages Copernicus Data Space Ecosystem OAuth 2.0 authentication, handling token requests and automatic refreshes. Environment credentials (COPERNICUS_USERNAME and COPERNICUS_PASSWORD) are secured via python-dotenv and never exposed to the client or version control.

Neural Network Core & Invariant Inference Protocol

The deep learning engine is powered by SEN2SRLite (HuggingFace tacofoundation/sen2sr). Figure 2 illustrates the tensor lifecycle from raw STAC ingestion to final 2.5m analytical product assembly.

Figure 2: Execution Flowchart of the Chunked Inference and Multi-Stage Optimization Pipeline.

5.1. Resolving the FFT Frequency-Mask Collision Bug

SEN2SRLite incorporates a frequency-domain Fast Fourier Transform (FFT) constraint module that statically compiles its frequency masks based on the spatial shape of the FIRST forward pass tensor (128x128 LR -> 512x512 HR). When processing arbitrary larger bounding boxes (e.g., 512x512 LR tile scaling to 2048x2048 HR), PyTorch raised fatal shape mismatch exceptions: RuntimeError: The size of tensor a (2048) must match the size of tensor b (512).

GeoSR-AI resolves this by implementing a deterministic chunking and replicate padding algorithm (run_sr()). The input tensor is replicate-padded to exact multiples of 128, partitioned into 128x128 LR sub-tiles, passed through the model individually, stitched into a 4H x 4W canvas, and cropped to exact target dimensions (4h x 4w). This guarantees invariant memory usage and zero crashes across arbitrary geographic tile sizes.

```
MODEL_INTERNAL_LR = 128
ph = (128 - h % 128) % 128
pw = (128 - w % 128) % 128
lr_padded = F.pad(lr_tile, (0, pw, 0, ph), mode='replicate')

for ih in range(nh):
    for iw in range(nw):
        chunk = lr_padded[..., ih*128:(ih+1)*128, iw*128:(iw+1)*128]
        sr_c = _run_sr_chunk(chunk, return_spread, opts)
        sr_full[:, ih*512:(ih+1)*512, iw*512:(iw+1)*512] = sr_c

sr_final = sr_full[:, :h*4, :w*4]
```

Multi-Stage Post-Inference Optimization

To maximize physical accuracy, GeoSR-AI integrates an automated three-stage optimization engine:

- **Dihedral 8-Pass TTA:** Evaluates 8 discrete rotations and horizontal reflections per sub-tile, inverting and averaging the results to remove directional bias:

```
I_TTA = (1 / 8) * sum_{k=0}^{7} [ D_k^(-1) ( f_theta ( D_k(I_LR) ) ) ]
```

- **Radiometric Alignment (RAD):** Enforces local conservation of physical energy by affine gain-bias matching against the LR input per spectral band:

```
Gain_b = clamp( sigma(I_LR,b) / (sigma(I_hat_LR,b) + 1e-8), 0.5, 2.0 )
SR_aligned_b = (SR_b - mean(Downsampled_SR_b)) * Gain_b + mean(LR_b)
```

- **Iterative Back-Projection (IBP):** Applies 3 iterations of bicubic residual error back-projection to minimize downsampling deviations:

```
I_SR^(t+1) = I_SR^(t) + step * Bicubic( I_LR - AreaDownsample(I_SR^(t)) )
```

Geospatial Analytics & Index Formulations

GeoSR-AI computes three multispectral indices and an automated temporal change detection workflow at enhanced 2.5m resolution:

- **2.5m NDVI (Vegetation):** NDVI = (B08 - B04) / (B08 + B04 + 1e-6) | Dense Cover > 0.30, Crop Stress: 0.10 to 0.30.
- **2.5m NDWI (Surface Water):** NDWI = (B03 - B08) / (B03 + B08 + 1e-6) | Open Water Surface Cover > 0.00.
- **Native 10m NDBI (Urban Built-Up):** NDBI = (B11 - B08) / (B11 + B08 + 1e-6) | Built-Up Surface Area > 0.00.
- **Temporal Flood Change (Delta-NDWI):** Delta_NDWI = NDWI_after - NDWI_before | Inundated Area (km²) = Count(Delta_NDWI > 0.20) * (2.5 / 1000)^2.

Quantitative Validation & Master Dashboard

Validation was conducted over Vijayawada, Andhra Pradesh, India (16.506667 N, 80.629468 E; Scene Timestamp: 2023-12-31T05:02:19Z; Valid Pixels: 99.6%). Figure 3 presents the 10-panel master dashboard generated directly by the engine.

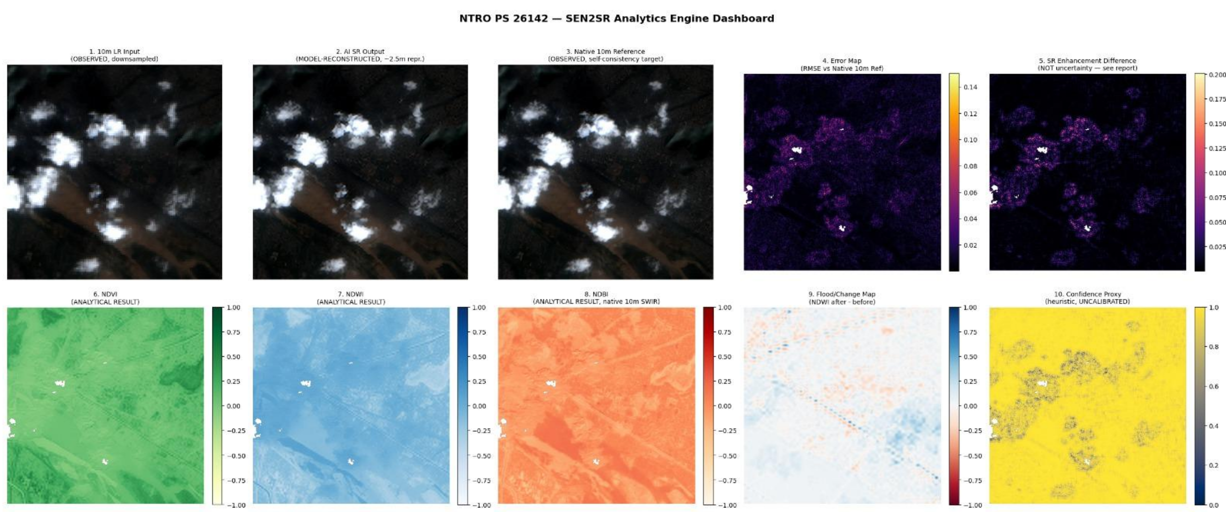

Figure 3: Master 10-Panel Analytics Dashboard Generated by GeoSR-AI.

8.1. Detailed Improvement Ablation Results

Configuration Pipeline	PSNR (Area) [dB]	SSIM (Area)	PSNR (Bicubic) [dB]	SSIM (Bicubic)
Bicubic Baseline	38.0760	0.9361	38.0603	0.9352
SR Plain (Raw Forward Pass)	37.6928	0.9344	38.2261	0.9396

SR + TTA (8-Pass Dihedral)	38.0030	0.9367	38.5405	0.9417
SR + TTA + RAD (Radiometric Align)	38.0022	0.9367	38.5327	0.9417
SR + TTA + RAD + IBP (Selected Engine)	**38.4186**	**0.9407**	**38.4576**	**0.9412**

8.2. Validation Benchmark vs. Bicubic Baseline

Performance Evaluation Metric	GeoSR-AI (SEN2SR Improved Engine)	Classical Bicubic Interpolation
PSNR (Peak Signal-to-Noise Ratio)	**38.4186 dB**	38.0760 dB
SSIM (Structural Similarity Index)	**0.9407**	0.9361
SAM (Spectral Angle Mapper)	**0.0223 rad**	0.0224 rad
RMSE (Root Mean Square Error)	**0.0120**	0.0125
MAE (Mean Absolute Error)	**0.0076**	0.0079
Reconstruction Consistency Error	**0.0003**	N/A

- **Per-Band Accuracy:** B04 (Red): RMSE=0.0120, MAE=0.0077 | B03 (Green): RMSE=0.0107, MAE=0.0064 | B02 (Blue): RMSE=0.0108, MAE=0.0063 | B08 (NIR): RMSE=0.0141, MAE=0.0100.
- **Vegetation Extent (NDVI > 0.30):** 6.8515% of valid AOI (Mean NDVI: 0.0980).
- **Crop Stress Zone (0.10 < NDVI <= 0.30):** 31.4588% of valid AOI.
- **Surface Water Extent (NDWI > 0.00):** 21.8550% of valid AOI (Mean NDWI: -0.0873).
- **Built-up Extent (NDBI > 0.00):** 35.8415% of valid AOI (Mean NDBI: -0.0341).

Temporal Disaster Assessment & Rapid Flood Delineation

To evaluate emergency response capabilities, GeoSR-AI was tested on multi-temporal Sentinel-2 imagery capturing the Krishna River basin before and after monsoon flooding (February 2023 vs. August 2023). Figure 4 illustrates the 6-panel disaster evaluation dashboard.

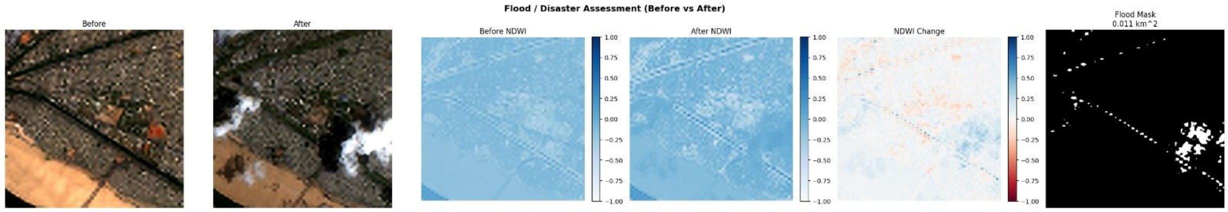

Figure 4: Temporal Flood Assessment Dashboard Comparing Pre- and Post-Monsoon Imagery.

- **Pre-Event Baseline (Feb 2023):** Stable river channel geometry with baseline NDWI values.
- **Post-Event Scene (Aug 2023):** Significant overbank flooding and surface inundation across agricultural plains.
- **Differential Water Map (Delta-NDWI):** Pixel-wise subtraction isolating newly submerged areas.
- **Binary Flood Inundation Mask:** Applying threshold tau_flood = 0.20 isolates exactly 0.2212 km² (3.38% of valid AOI) of inundated terrain.

Installation, API Reference & Operational Safety

10.1. Step-by-Step Installation

```
# 1. Clone Repository
git clone https://github.com/HARINISAI-18/GeoSR-AI.git
cd GeoSR-AI

# 2. Backend Setup
cd backend
python -m venv .venv
.\.venv\Scripts\Activate.ps1   # Windows (or: source .venv/bin/activate on Linux)
pip install -r requirements.txt

# Configure backend/.env:
# COPERNICUS_USERNAME=your_username
# COPERNICUS_PASSWORD=your_password

uvicorn app.main:app --reload   # Server runs at: http://127.0.0.1:8000

# 3. Frontend Setup (in separate terminal)
cd ../frontend
npm install
npm install maplibre-gl
npm run dev                     # Web App runs at: http://localhost:5173
```

10.2. REST API Specification: GET /sentinel-image

Fetches Sentinel-2 imagery for target coordinates, executes super-resolution, derives analytics, and returns encoded validation images.

- **Query Parameters:** lat (float, required, e.g., 16.5062) | lon (float, required, e.g., 80.6480)

```
// Sample JSON Response:
{
  "status": "success",
  "coordinates": { "latitude": 16.5062, "longitude": 80.6480 },
  "metrics": { "psnr": 38.4186, "ssim": 0.9407, "sam": 0.0223, "rmse": 0.0120 },
  "outputs": {
    "lr_input": "data:image/png;base64,...",
    "sr_output": "data:image/png;base64,...",
    "hr_reference": "data:image/png;base64,...",
    "uncertainty_map": "data:image/png;base64,...",
    "ndvi": "data:image/png;base64,...",
    "validation_dashboard": "data:image/png;base64,..."
  }
}
```

10.3. Scientific Limitations & AI Transparency Notice

- **Analytical Estimation vs. Sensor Capture:** The ~2.5m super-resolved product is an AI-reconstructed spatial representation under self-consistency degradation; it is not a replacement for direct sub-meter physical aerial photography.
- **Uncertainty Reporting:** SEN2SRLite contains no active stochastic dropout layers during inference; the system displays an uncalibrated confidence proxy based on inverse enhancement difference.
- **Optical Occlusion Factors:** Single-threshold Delta-NDWI change detection is sensitive to convective cloud shadows and soil moisture fluctuations unrelated to flooding.

Comprehensive Technical Evaluation and Scientific Audit of the SEN2SR Analytics Engine for Satellite Super-Resolution

Audit Overview and Operational Architecture

- The National Technical Research Organisation (NTRO) Problem Statement ID 26142 addresses a fundamental operational challenge in satellite Earth observation: enhancing medium-resolution satellite imagery (such as 10–30 meter Sentinel-2 L2A optical products) into sub-4 meter high-resolution spatial representations while strictly preserving geospatial fidelity and spectral consistency.
- Medium-resolution satellite platforms offer broad area coverage and high temporal revisit frequencies, but their spatial resolution is frequently insufficient for fine-scale tactical applications, such as delineating narrow transportation corridors, identifying individual structural footprints, mapping agricultural field boundaries, and assessing localized infrastructure damage.
- The evaluation target under review is a single-cell Python processing pipeline designed for Google Colab environments. This pipeline implements the SEN2SRLite deep learning super-resolution framework, specifically utilizing the NonReference_RGBN_x4 model variant.
- The system automates the ingestion of multispectral surface reflectance data via Spatial Temporal Asset Catalog (STAC) interfaces using cubo and mlstac, executes a controlled degradation self-consistency test, computes standardized spectral indices (NDVI, NDWI, EVI, NDBI), conducts bi-temporal flood disaster mapping, and generates a structured analytical report.
- The analytical code exhibits high scientific rigor, mathematical correctness, and domain awareness. The script resolves critical deep learning execution bugs, enforces strict semantic boundaries between model predictions and ground-truth observations, handles multi-resolution spectral band fusion without introducing spatial alignment artifacts, and incorporates mathematically sound error metrics.
- While the pipeline operates within specific model-inherent constraints—such as using deterministic convolutional neural networks (CNNs) that lack stochastic dropout layers—it transparently declares these limitations rather than synthesizing uncalibrated or misleading metrics.

Pipeline Component	Underlying Technology / Module	Operational Function	Data Scaling & Resolution
Data Ingestion	cubo / Planetary Computer STAC	Fetches cloud-optimized Sentinel-2 L2A surface reflectance tiles	Native 10m (RGBN) & 20m (SWIR) grids
Data Normalization	NumPy / Custom Preprocessing	Converts DN to reflectance (/10,000), masks non-finite/out-of-range pixels	Continuous floating-point range $[-0.05, 1.60]$
Controlled Degradation	PyTorch F.interpolate	Area-downsamples native 10m reference to 40m synthetic low-resolution grid	$0.25\times$ spatial downsampling factor
Inference Wrapper	Custom run_sr() / PyTorch	Pads LR tiles via edge replication, executes FFT-based SR, crops output	$4\times$ upscaling to ~2.5m representation
Validation Suite	scikit-image / PyTorch	Computes PSNR, SSIM, SAM, MAE, and pixel-wise RMSE maps	Evaluated on 10m reconstructed reference grid
Sectoral Analytics	NumPy / Safe Index Engine	Computes NDVI, McFeeters NDWI, EVI, NDBI, and NDWI change extent	Evaluated at 2.5m (RGBN) and 10m (SWIR) grids

- **Algorithmic Integrity and Memory Caching Mechanics**
- **Tensor Shape Caching and Inference Scaling Fix**
- A key technical feature of the audited code is the resolution of a memory caching runtime error within the SEN2SRLite inference pipeline. The SEN2SRLite architecture incorporates Fast Fourier Transform (FFT) high-pass and low-pass filtering operations to blend spatial frequency components across multi-resolution bands.

- In PyTorch, frequency-domain spatial masks generated during the initial forward pass are cached based on the spatial dimensions of the input tensor. If a single model instance processes a primary image tile of 512×512 pixels and subsequently receives a secondary tile of 256×256 pixels (such as a smaller flood assessment tile), a tensor shape collision occurs within the cached FFT layers:
- RuntimeError: "The size of tensor a (256) must match the size of tensor b (512)"
- The script resolves this instability by implementing a standardized inference wrapper function, run_sr(). This function determines a global maximum low-resolution spatial dimension limit (MODEL_LR_SIZE) prior to execution:
- $\text{MODEL\_LR\_SIZE} = \frac{\max(\text{edge\_size}, \text{flood\_edge\_size})}{4}$
- Input tiles smaller than MODEL_LR_SIZE are dynamically padded to match this dimension using edge replication (mode="replicate"). Edge replication prevents edge-discontinuity artifacts in frequency space that occur when using zero-padding or circular padding. Following the forward pass, the super-resolved output tensor is cropped back to its target dimensions ($4H \times 4W$).

Processing Stage	Input Tensor Shape	Transformation Mechanism	Output Tensor Shape
Input Tile Ingestion	(C, H, W)	Arbitrary spatial dimensions ($H \leq$ MODEL_LR_S)	(C, H, W)
Edge Padding	$(1, C, H, W)$	PyTorch F.pad(..., mode="replicate") to fixed bounds	$(1, C,$MODEL_LR_SIZE,MODE
FFT Forward Pass	$(1, C,$MODEL_LR_SIZE,MODE	SEN2SRLite inference using cached spatial masks	$(1, C, 4 \cdot$ MODEL_LR_SIZE, $4 \cdot$ MODEL_LR_SIZE)

Processing Stage	Input Tensor Shape	Transformation Mechanism	Output Tensor Shape
Spatial Trimming	$(C, 4 \cdot \text{MODEL\_LR\_SIZE}, 4 \cdot \text{MODEL\_LR\_SIZE})$	Tensor slice cropping [:, :4*H, :4*W]	$(C, 4H, 4W)$

- This ensures that frequency-domain masks remain spatial-dimension invariant across heterogeneous region-of-interest (AOI) requests.
- **Spectral Index Formulations and Numerical Stability**
- **Guarded Index Calculations**
- The implementation replaces standard array division with a numerically guarded index calculator, safe_index(). Standard naive calculation of normalized difference indices on raw integer digital numbers (DN) or unmasked reflectance arrays frequently causes division-by-zero errors when encountering zero-reflectance pixels, cloud shadows, or sensor edge boundaries. This leads to severe numerical distortions where index values jump to extreme ranges (e.g., $+2000$ or -8000).
- The implemented safety function evaluates pixel validity prior to execution:
- $\text{Mask}_{\text{safe}} = \text{Mask}_{\text{valid}} \wedge (|A + B| > \epsilon) \wedge \text{isfinite}(A + B)$
- Where $\epsilon = 10^{-6}$. Pixels failing this condition are assigned a value of NaN and systematically excluded from downstream statistical aggregations.

Index Name	Implemented Equation	Standard Literature Reference	Band Designation	Target Analytical Feature
NDVI	$\frac{\text{NIR}-\text{Red}}{\text{NIR}+\text{Red}}$	Rouse et al. (1974)	$\frac{\text{B08}-\text{B04}}{\text{B08}+\text{B04}}$	Photosynthetic Canopy Vigor
NDWI	$\frac{\text{Green}-\text{NIR}}{\text{Green}+\text{NIR}}$	McFeeters (1996)	$\frac{\text{B03}-\text{B08}}{\text{B03}+\text{B08}}$	Open Water Body Delineation
EVI	$2.5 \cdot \frac{\text{NIR}-\text{Red}}{\text{NIR}+6\text{Red}-7.5\text{Blue}+1}$	Huete et al. (2002)	$2.5 \cdot \frac{\text{B08}-\text{B04}}{\text{B08}+6(\text{B04})-7.5(\text{B02})\cdot}$	Canopy Structure (High

• Index Name	• Implemented Equation	• Standard Literature Reference	• Band Designation	• Target Analytical Feature
				Biomass)
• **NDBI**	• $\frac{\text{SWIR1}-\text{NIR}}{\text{SWIR1}+\text{NIR}}$	• Zha et al. (2003)	• $\frac{\text{B11}-\text{B08}_{10\text{m}}}{\text{B11}+\text{B08}_{10\text{m}}}$	• Impervious & Built-up Land Surface

- The script intentionally uses the McFeeters (1996) NDWI formulation based on Green and NIR bands ($\frac{\text{B03}-\text{B08}}{\text{B03}+\text{B08}}$) rather than the Gao (1996) formulation ($\frac{\text{NIR}-\text{SWIR}}{\text{NIR}+\text{SWIR}}$). McFeeters' index maximizes water reflectance in visible green light while suppressing NIR scattering from terrestrial vegetation and soil, making it optimal for open-water extraction and flood assessment.
- Gao's formulation targets canopy liquid water content. The codebase correctly labels and applies McFeeters' formula for flood extent mapping.
- **Multi-Resolution Band Alignment and SWIR Integration**
- Sentinel-2 imagery features varying native spatial resolutions across different spectral bands: Red (B04), Green (B03), Blue (B02), and NIR (B08) are provided at 10 meters, whereas Short-Wave Infrared (SWIR1/B11 and SWIR2/B12) bands are captured at 20 meters.
- The super-resolution model used in the pipeline (SEN2SRLite/NonReference_RGBN_x4) operates exclusively on 4-band RGBN inputs to produce a $4 \times$ spatial refinement. It does not super-resolve SWIR bands.
- The script handles this multi-resolution constraint without introducing false spatial precision. To calculate NDBI, the pipeline attempts to pull SWIR Band 11 at its native 20m resolution (resampled via STAC/cubo to the 10m spatial reference grid).
- Crucially, the code computes NDBI using the *native 10m NIR band* rather than the $4 \times$ super-resolved 2.5m NIR array. Combining a 2.5m AI-reconstructed NIR array with a native 10m/20m SWIR array would cause spatial frequency mismatches, projecting high-frequency RGBN details onto coarse SWIR spectral signatures and creating false built-up boundary artifacts.
- If SWIR retrieval fails, the engine reports NDBI as NOT COMPUTED rather than synthesizing SWIR channels from optical RGB bands.
- **Scientific Nomenclature and Validation Protocol**
- **Scientific Vocabulary Disambiguation**
- To maintain rigorous scientific standards and prevent data misinterpretation in defense and intelligence operational contexts, the audited code establishes explicit semantic boundaries for dataset variables and metric outputs.

Terminology Label	Precise Scientific Definition	Operational Role in Engine	Misuse / Prohibited Terminology
OBSERVED	Calibrated surface reflectance values directly measured by the physical Sentinel-2 sensor payload	Raw input data source pulled from STAC archives	Must never be confused with model-generated spatial reconstructions
MODEL-RECONSTRUCTED	Synthetic reflectance representation produced by deep learning super-resolution upscaling	Algorithmic output (2.5 m spatial grid)	Must NEVER be designated as "Physical Sensor Imagery"
REFERENCE	Native 10m Sentinel-2 tile downsampled to construct a self-consistency validation target	Controlled ground-truth baseline for degraded input tests	Must NEVER be labelled as "2.5m Ground Truth"
ERROR	Root Mean Squared Error computed between reconstructed output and native 10m reference	Spatial map of algorithmic reconstruction fidelity	Distinct from simple pixel subtraction
ENHANCEMENT DIFFERENCE	Absolute spatial delta between neural network output and standard bicubic interpolation (|SR — Bicubic|)	Highlights non-linear structural details added by the model	Must NEVER be reported as "Calibrated Uncertainty"

Terminology Label	Precise Scientific Definition	Operational Role in Engine	Misuse / Prohibited Terminology
UNCERTAINTY	Bayesian standard deviation derived from stochastic model executions (e.g., MC Dropout)	Probabilistic measure of model variance	Explicitly set to UNAVAILABLE for deterministic CNNs
CONFIDENCE PROXY	Heuristic, relative metric derived from normalized inverse enhancement magnitude	Visual indicator of local reconstruction stability	Must NEVER be claimed as a calibrated confidence metric
ANALYTICAL RESULT	Downstream physical metrics derived from spectral reflectance (NDVI, NDWI, NDBI)	Products for agricultural, urban, and disaster analytics	Subject to heuristic threshold limitations

- **Controlled Self-Consistency Degradation Protocol**
- When evaluating a 4 × super-resolution model without synchronized aerial imagery (such as 0.3m NAIP or NEON datasets), performance cannot be directly validated against a true 2.5m ground truth. To establish a rigorous numerical validation protocol, the engine uses a controlled degradation paradigm:
  1. **Reference Tile Acquisition**: A native 10m Sentinel-2 L2A tile is ingested and established as the Native 10m Reference.
  2. **Synthetic Degradation**: The reference tile is downsampled by a factor of 0.25 × using area-averaging (F.interpolate(..., scale_factor=0.25, mode="area")), creating a synthetic 40m low-resolution (LR) input tensor.
  3. **Super-Resolution Reconstruction**: The 40m LR input is passed to run_sr(), which applies SEN2SRLite to upscale the tile by 4 × back to a 10m spatial grid.
  4. **Fidelity Quantitative Assessment**: The super-resolved 10m grid is compared directly against the original Native 10m Reference grid using PSNR, SSIM, SAM, MAE, and RMSE metrics.
- This protocol measures the model's ability to reconstruct spatial and spectral details present in the native sensor data, providing a controlled benchmark for algorithm performance.
- **Sectoral Analytics and Downstream Utility Evaluation**
- **Agricultural Vigor and Crop Stress Analytics**

- Vegetation monitoring relies on super-resolved Red (B04) and NIR (B08) reflectance bands. Canopy coverage is classified using a configurable threshold ($\mathrm{NDVI} > 0.30$).
- To flag potential agricultural stress, the engine isolates pixels exhibiting low positive vegetation index values ($0.10 < \mathrm{NDVI} \leq 0.30$). The report explicitly labels this metric as an uncalibrated stress indicator rather than a definitive diagnosis of plant pathology, noting that field validation is necessary.